BioInteract: Interpretable Drug–Target Interaction Prediction via Residue-Level Cross-Attention with Biological Prior Knowledge

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ABSTRACT

Accurate prediction of drug–target interactions (DTIs) can support the prioritisation of candidate pairs, yet many deep-learning models provide little visibility into the features used for prediction. We present BioInteract, a DTI framework combining a chemistry-aware molecular graph encoder, an ESM-2/physicochemical protein residue encoder with a configured domain-label channel, and bidirectional cross-attention. The model produces atom–residue attention matrices and atom-level Grad-CAM scores as model-native attribution outputs. A full-precision reconstruction from the released random-split checkpoint and inputs achieved an AUROC of 0.904 on the Davis kinase inhibitor benchmark; the corresponding entity-held-out reconstructions are reported with their complete provenance. We additionally audit chemical and sequence similarity across these partitions and evaluate exact-sequence-grouped cold-start protocols. The attribution analyses generate hypotheses about model-relevant atoms and residues, but do not constitute predicted three-dimensional contacts or structural mechanisms. The released input package contains no curated per-protein domain annotations, so the domain-label channel uses its unknown label and is not treated as domain evidence. BioInteract therefore provides a transparent starting point for prioritisation and for experimentally testable hypotheses, subject to the dataset and structural-validation limitations stated below.

Keywords

Drug–target interaction; interpretable deep learning; cross-attention; graph neural network; protein language model; model attribution

1 Introduction

The emergence of "AI for Science" has increased expectations that computational models should not merely maximise predictive accuracy, but should also expose the evidence and uncertainty needed for scientific hypothesis generation.[1] This expectation is especially important in drug discovery, where identifying drug–target interactions (DTIs) requires reliable prioritisation together with inspectable outputs that can guide, but not replace, mechanistic follow-up.

Experimental profiling of binding affinities through surface plasmon resonance, isothermal titration calorimetry, or kinase activity assays is laborious and prohibitively expensive. The Davis kinase inhibitor dataset[2] alone required systematic measurement of 72 inhibitors against 442 kinases—a campaign spanning only a fraction of the druggable proteome. Computational DTI prediction can ease this bottleneck by prioritising the most promising candidates and thereby reducing cost and attrition in development pipelines.[3,4] Yet the field has largely remained trapped in a *predictive-accuracy race*: successive deep learning architectures—DeepDTA,[5] GraphDTA,[6] MolTrans,[7] DrugBAN[8]—have incrementally improved benchmark metrics while leaving an important modelling question only partly addressed: *which input features drive a given prediction?*

This question lies at the heart of the broader "AI for Science" agenda. A prediction with no interpretable output is, at best, a hypothesis generator. Moving from *prediction* to hypothesis generation therefore benefits from models that expose internal attribution patterns—for example, attention distributed across drug atoms and protein residues and gradient-based atom scores—rather than returning only a scalar probability. These outputs can prioritise follow-up experiments, but they are not direct measurements of physical contacts or pocket geometry.

Classical DTI methods illuminate the fundamental tension between interpretability and accuracy. Structure-based approaches (molecular docking, molecular dynamics) provide structural inference anchored in three-dimensional protein geometry,[9] yet are constrained by the availability of high-quality crystal structures and the steep computational cost of free-energy calculations. Ligand-based methods (QSAR, molecular fingerprints)[10] circumvent structural requirements at the expense of mechanistic transparency. Neither paradigm fully satisfies the twin demands of modern drug discovery.

Deep learning has transformed predictive capabilities in this domain. Pretrained protein language models such as ESM-2[11] encode evolutionary and structural information at residue resolution, while graph neural networks faithfully capture molecular topology. Concurrently, chemical language models have demonstrated that pretrained molecular representations support accurate and interpretable prediction of reactivity and toxicity,[12] and chemistry-enhanced transformer architectures such as Radical-Net recover atom-level reaction mechanisms.[13] Together, these advances underscore the importance of domain-specific pretraining and chemistry-aware inductive biases in molecular AI;[14] their integration into interpretable DTI frameworks, however, has remained limited.

In this study, "biological prior knowledge" refers specifically to the sequence information encoded by frozen ESM-2 pretraining together with explicit per-residue physicochemical descriptors. It does not denote curated Pfam, InterPro, binding-site, or other protein-domain annotations.

The "AI for Science" paradigm benefits from attribution mechanisms that are part of the prediction pathway rather than appended only after training. Attention visualisation and SHapley Additive exPlanations (SHAP) can nevertheless have faithfulness limitations.[15] Recent successes in other scientific domains illustrate the power of built-in interpretability: attention-based models have elucidated environmental drivers of algal blooms,[16] discriminative cross-category attention networks capture fine-grained spatial–temporal correspondences in action recognition,[17] and uncertainty-guided spatial–temporal learning resolves subtle cross-modal dependencies in skeleton-based recognition.[18] A unifying principle emerges: *when attention constitutes an integral component of the prediction pathway, the resulting attributions can be examined consistently with model output, but still require external validation before mechanistic interpretation.*

Motivated by these observations, we present **BioInteract**, a framework that brings model-native attribution outputs to binary high-affinity DTI classification. BioInteract rests on three architectural components:

1. **Chemistry-aware molecular graph encoding.** Drug molecules are represented as attributed graphs encoding functional-group-relevant properties (H-bond donor/acceptor status, hydrophobicity, partial charges) that are available to the learned molecular representation.

2. **Protein residue encoding with a disclosed domain-label limit.** Protein representations fuse ESM-2 residue embeddings with per-residue physicochemical properties and a configured domain-label channel. No curated per-protein domain-annotation files are present in the released Davis input package; the channel therefore receives the unknown label at every position and is not interpreted as Pfam/InterPro information.

3. **Bidirectional cross-attention attribution module.** A multi-head cross-attention mechanism computes a matrix $\mathbf{M} \in \mathbb{R}^{n_{\text{atoms}} \times n_{\text{residues}}}$ of learned attention weights between drug-atom and protein-residue representations. The matrix is an integral component of the forward pass; it is used here as a model-native attribution output, not as a predicted atom-level structural contact map.

We evaluate BioInteract on the Davis kinase inhibitor dataset[2] under a random pair split and two entity-held-out protocols: Drug-ID-held-out and Target-ID-held-out. We report chemical and sequence-similarity diagnostics for these partitions, because identifier-level exclusion alone does not guarantee scaffold- or sequence-level novelty. We also add exact-sequence-grouped cold-target and exact-sequence-grouped cold-both evaluations. The attribution analyses—cross-attention maps, Grad-CAM scores, attention-concentration summaries, and case studies—are interpreted as model behaviour rather than as independent evidence of molecular recognition mechanisms.

2 Related Work

2.1 Deep Learning for DTI Prediction

Deep learning approaches to DTI prediction have evolved through several generations.[19] *Sequence-based* models encode both drug SMILES and protein amino acid sequences as one-dimensional strings. DeepDTA[5] employs parallel one-dimensional convolutional neural networks (CNNs) for drug and protein encoding; WideDTA[20] extends this framework with additional molecular descriptors. Although computationally efficient, these models discard molecular topology and are limited in their capacity to model long-range intramolecular interactions.

Graph-based models overcome this limitation by representing drugs as attributed molecular graphs. GraphDTA[6] benchmarks a graph convolutional network (GCN), graph attention network (GAT), graph isomorphism network (GIN), and hybrid GAT–GCN encoder while retaining a CNN for proteins. MolTrans[7] introduces mutual interaction learning between molecular substructure representations and protein subsequences.

Attention-based models include architectures that learn local or cross-modal representations for DTI/DTA prediction. TransformerCPI[21] applies transformer attention to drug–protein interaction patterns. DrugBAN[8] introduces a dual bilinear attention network for fine-grained local interaction modelling. AttentionDTA[22] employs multi-head self-attention for both drug and protein representations. In most prior work, however, attention operates as a computational device rather than as an interpretable output channel.

2.2 Pretrained Protein Language Models

ESM-2,[11] trained on millions of protein sequences via masked language modelling, produces residue-level embeddings that encode evolutionary conservation, secondary structure propensity, and contact information. These pretrained representations substantially outperform hand-crafted features on downstream tasks.[23] In DTI prediction, pretrained representations have already been combined with graph drug encoders and interaction modules. For example, GEFA uses pretrained protein representations in an early-fusion graph architecture,[24] PGraphDTA replaces a CNN target encoder with protein language-model features,[25] and HCAF-DTA combines protein representations with bidirectional cross-attention.[26] BioInteract therefore does not claim to be the first application of a protein language model, a graph encoder, or cross-attention to DTI/DTA prediction. Its specific contribution is the submitted integration of a chemistry-aware molecular graph encoder, residue-level ESM-2/physicochemical features with a configured domain-label channel, and inspectable model-native attribution outputs for the Davis binary-classification setting. These outputs are not treated as biologically grounded structural evidence without independent validation.

2.3 Interpretability in DTI Models

Existing interpretability approaches include post-hoc methods (attention-weight visualisation, gradient-based attribution, and SHAP values[27]) and model-native attribution mechanisms. Attention weights do not necessarily equal feature importance.[15] BioInteract uses a cross-attention matrix that contributes to prediction and can be inspected as a model-native attribution. It does not by itself establish atom-level contacts, binding modes, or physical interaction types. In contrast, models such as MONN use explicit non-covalent interaction supervision derived from structural complexes;[28] no comparable supervision is used here.

2.4 Cold-Start Evaluation

Random-split evaluation permits information leakage through shared drugs or targets between training and test sets, yielding inflated estimates.[29] Cold-start protocols address this by ensuring that test identifiers are absent from training data. We report the original Drug-ID-held-out and Target-ID-held-out protocols alongside the random split, and separately use exact-sequence grouping for stricter within-Davis stress tests.

3 Methods

3.1 Problem Formulation

Given a drug molecule d represented by its SMILES string and a target protein t represented by its amino acid sequence, we formulate DTI prediction as binary classification: predict whether d binds t with high affinity. Following the established protocol for the Davis dataset,[2] continuous K_d values are binarised at 30 nM ($pK_d > 7.52$), classifying pairs with $K_d < 30\,\text{nM}$ as positive interactions.

3.2 Architecture Overview

BioInteract comprises five sequential modules: (1) a graph isomorphism network with edge features (GINE)-based drug encoder producing per-atom representations, (2) a target encoder combining ESM-2 embeddings with physicochemical features and a configured domain-label channel, (3) a bidirectional cross-attention module computing attention matrices between ligand-atom and protein-residue representations, (4) gated attention pooling to aggregate variable-length representations, and (5) a multilayer perceptron (MLP) prediction head. The full model contains 2,442,083 trainable parameters.

3.3 Drug Molecular Graph Encoder

3.3.1 Chemistry-Aware Atom Featurisation

Each atom is represented by a 52-dimensional feature vector encoding:

- **Topological features** (38 dimensions): atom type (one-hot, 20 types), hybridisation state (6), formal charge (6), number of attached hydrogens (6).

- **Structural features** (9 dimensions): degree, aromaticity, ring membership, and ring size membership (3–8).
- **Functional features** (5 dimensions): hydrogen-bond donor and acceptor status, Crippen logP contribution, molar refractivity, and Gasteiger partial charge.

Bond features (16 dimensions) encode bond type, conjugation, ring membership, and stereochemistry.

3.3.2 GINE Message Passing

We employ a 3-layer GINE architecture:[30]

$$\mathbf{h}_v^{(\ell+1)} = \mathrm{MLP}^{(\ell)}\left((1+\varepsilon^{(\ell)})\,\mathbf{h}_v^{(\ell)} + \sum_{u\in\mathcal{N}(v)} \mathrm{ReLU}\left(\mathbf{h}_u^{(\ell)} + \mathbf{e}_{vu}\right)\right) \tag{1}$$

where $\mathbf{h}_v^{(\ell)}$ is the representation of atom v at layer ℓ, $\mathbf{e}_{vu}$ denotes the edge feature vector, and $\varepsilon^{(\ell)}$ is a learnable scalar. Each layer incorporates batch normalisation, ReLU activation, dropout ($p = 0.2$), and a residual connection; the hidden dimension is $D = 256$. Global graph pooling is *not* applied at this stage, preserving individual atom representations for the subsequent cross-attention step.

The reported final training runs use the fixed molecular graphs produced by the stated featurisation pipeline. The final scripts set both DropNode and DropEdge probabilities to zero, so no stochastic graph augmentation is applied during training or evaluation.

3.4 Protein Target Encoder

3.4.1 ESM-2 Residue Embeddings

Residue-level representations are extracted from ESM-2 (150 M parameters, `esm2_t30_150M_UR50D`),[11] yielding 640-dimensional embeddings for each amino acid. These embeddings encode evolutionary conservation, secondary structure propensity, and long-range contact information learned from 65 million protein sequences. ESM-2 parameters are frozen throughout training.

3.4.2 Physicochemical Property Encoding

Each residue is augmented with a 4-dimensional physicochemical vector:

$$\mathbf{p}_i = \left[\mathrm{hydrophobicity}_i,\ \mathrm{mol.\ weight}_i,\ \mathrm{charge}_i,\ \mathrm{polarity}_i\right] \tag{2}$$

Hydrophobicity follows the Kyte–Doolittle scale.[31] These values provide residue-level physicochemical descriptors; by themselves, they do not identify pocket membership, solvent exposure, or interaction partners.

3.4.3 Configured Domain-Label Channel and Availability

The implementation contains a learnable domain-label embedding layer:

$$\mathbf{d}_i = \mathrm{DomainEmbed}(\mathrm{domain\_type}_i) \in \mathbb{R}^{32} \tag{3}$$

The label builder assigns the unknown `NONE` type when a per-protein annotation file is unavailable. The released Davis package contains no such annotation files, so all archived input positions use this unknown label. We therefore do not represent this channel as curated Pfam/InterPro context, as a binding-site label, or as evidence for a domain-specific conclusion.

3.4.4 Feature Fusion

The three feature sources are concatenated and projected through a shared linear transformation:

$$\mathbf{r}_i = \mathrm{Proj}([\mathbf{e}_i \,\|\, \mathbf{p}_i \,\|\, \mathbf{d}_i]) \in \mathbb{R}^D \tag{4}$$

where $\mathbf{e}_i \in \mathbb{R}^{640}$ is the ESM-2 embedding and Proj is a two-layer MLP with layer normalisation and Gaussian error linear unit (GELU) activation. In the released configuration, $\mathbf{d}_i$ is the shared unknown domain-label embedding rather than a curated per-residue annotation. The projected dimension $D = 256$ matches the drug encoder output.

3.5 Cross-Attention Interaction Module

The cross-attention module produces the model-native attribution used in this study. It computes bidirectional attention between ligand-atom and protein-residue representations, producing an explicit atom–residue attention matrix.

3.5.1 Drug-to-Protein Attention

$$\mathrm{Attn}(\mathbf{Q}_d, \mathbf{K}_p, \mathbf{V}_p) = \mathrm{softmax}\left(\frac{\mathbf{Q}_d \mathbf{K}_p^\top}{\sqrt{d_k}}\right)\mathbf{V}_p \tag{5}$$

where $\mathbf{Q}_d = \mathbf{H}_d \mathbf{W}_Q^d$, $\mathbf{K}_p = \mathbf{R}\mathbf{W}_K^p$, $\mathbf{V}_p = \mathbf{R}\mathbf{W}_V^p$, with $H = 8$ heads and head dimension $d_k = D/H = 32$.

3.5.2 Protein-to-Drug Attention

Symmetrically, protein residues attend over drug atoms:

$$\mathrm{Attn}(\mathbf{Q}_p, \mathbf{K}_d, \mathbf{V}_d) = \mathrm{softmax}\left(\frac{\mathbf{Q}_p \mathbf{K}_d^\top}{\sqrt{d_k}}\right)\mathbf{V}_d \tag{6}$$

3.5.3 Attribution Map Extraction

The atom–residue attribution map is obtained by averaging drug-to-protein attention weights across all heads:

$$\mathbf{M} = \frac{1}{H}\sum_{h=1}^{H} \mathrm{Attn}_h(\mathbf{Q}_d, \mathbf{K}_p) \in \mathbb{R}^{n \times L} \tag{7}$$

Because $\mathbf{M}$ is computed within the forward pass rather than post-hoc, it is a model-native attribution matrix. It is not interpreted as a physical contact map or as an assignment of a specific interaction type to an atom pair.

3.6 Gated Pooling and Prediction

Variable-length representations are aggregated via gated attention pooling:

$$\mathbf{z} = \sum_i \alpha_i \cdot g(\mathbf{x}_i), \qquad \alpha_i = \frac{\exp(f(\mathbf{x}_i))}{\sum_j \exp(f(\mathbf{x}_j))} \tag{8}$$

where $f\colon \mathbb{R}^D \to \mathbb{R}$ and $g\colon \mathbb{R}^D \to \mathbb{R}^D$ are learned two-layer MLPs with Tanh activation. The pooled drug and protein representations are concatenated and passed through a two-hidden-layer MLP ($256 \to 128$; ReLU; dropout $p = 0.3$) followed by sigmoid activation to produce the binding probability.

3.7 Training Procedure

The Davis dataset is severely class-imbalanced ($\approx$5 % positive rate). We address this with weighted binary cross-entropy:

$$\mathscr{L} = -\frac{1}{N}\sum_{i=1}^{N}\left[w_+ \cdot y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)\right] \tag{9}$$

with $w_+ = N_{\mathrm{neg}}/N_{\mathrm{pos}} \approx 18$, combined with label smoothing ($\varepsilon = 0.05$). Optimisation uses AdamW[32] ($\eta = 10^{-4}$, weight decay $\lambda = 10^{-5}$) with cosine annealing and a 5-epoch linear warmup. Early stopping monitors validation AUROC (patience = 15, maximum 100 epochs), and the decision threshold is selected by maximising F1 on the validation precision–recall curve.

3.8 Interpretability Analysis

3.8.1 Residue-Level Attention Profiling

For a given drug–target pair, per-residue attention scores are obtained by summing the interaction map over all drug atoms:

$$s_j = \sum_{i=1}^{n} M_{ij}, \quad j = 1, \ldots, L \tag{10}$$

These values are model-attribution scores and are not assigned a biological hotspot label.

3.8.2 Grad-CAM Functional-Group Attribution

Grad-CAM[33] is adapted to the graph-neural-network (GNN) setting:

$$\mathrm{CAM}_v = \mathrm{ReLU}\left(\sum_k w_k \cdot A_v^k\right), \quad w_k = \frac{1}{|\mathscr{V}|}\sum_v \frac{\partial z}{\partial A_v^k} \tag{11}$$

where A_v^k is channel k of the third (final) GINE message-passing layer and z is the pre-sigmoid positive-class logit. Gradients are therefore taken with respect to the logit, not the post-sigmoid probability. Functional groups are detected with predefined SMARTS patterns (aromatic ring, hydroxyl, amino, amide, carboxyl, sulfonyl, halogen, ether, heterocycle N, and carbonyl). For each group, all RDKit substructure matches are merged into a deduplicated atom set and the group score is the mean of the normalised atom-level scores over that set. An atom can match more than one group pattern. These scores indicate sensitivity of the prediction to the learned representation and do not identify chemical bonds or contacts in a three-dimensional complex.

3.8.3 Attention Sparsity

For each drug–target pair, residue scores are normalised by that pair's maximum:

$$\widetilde{s}_j = \frac{s_j}{\max_k s_k}. \tag{12}$$

Pair-level sparsity is then

$$\text{Sparsity} = \frac{|\{j : \widetilde{s}_j < 0.1\}|}{L}. \tag{13}$$

The reported global value uses all 1,506 positive Davis records, rather than a single test partition, evaluated with the archived `checkpoints/best.pt` checkpoint. Each pair is normalised separately; the 1,301,380 valid-residue scores are then concatenated before the global distribution and fraction are computed. Consequently, all original train/validation/test memberships are represented and longer valid target sequences contribute more residue-level observations.

4 Experimental Setup

4.1 Dataset

The Davis kinase inhibitor dataset[2] comprises 68 drugs (kinase inhibitors) and 442 targets (protein kinases) forming 30,056 drug–target pairs. Binding affinities (K_d) are binarised at 30 nM, yielding approximately 5 % positive interactions.

4.2 Data Splitting Protocols and Similarity Audit

- **Random split** (70/10/20): drug–target pairs are assigned randomly; shared entities may appear across splits.
- **Drug-ID-held-out split**: drugs are partitioned such that all interactions of a given compound identifier appear in exactly one split. This prevents drug-identifier overlap but does not impose a scaffold-similarity threshold.
- **Target-ID-held-out split**: targets are partitioned analogously; test target identifiers are absent from training. This initial entity-held-out protocol does not impose a sequence-similarity threshold.

For the Drug-ID-held-out audit, we computed non-chiral radius-2, 1024-bit Morgan fingerprints with RDKit and reported the maximum Tanimoto similarity between each test compound and the training compounds. For the Target-ID-held-out audit, we used RapidFuzz `fuzz.ratio`, divided by 100, where normalised Indel similarity is $1 - d_{\text{Indel}}/(|x|+|y|)$ and substitutions are represented by one deletion plus one insertion. We computed the maximum value between each test sequence and all training sequences and separately identified exact duplicate sequences. The original Target-ID-held-out partition contained six exact-sequence groups crossing the training and test sets; it is therefore not described as a sequence-novel evaluation. We additionally constructed exact-sequence-grouped cold-target and exact-sequence-grouped cold-both partitions, assigning all target identifiers with an identical amino-acid sequence to the same split. These reviewer-requested protocols use the same Davis records, 70/10/20 partition ratios, and seed 42.

For the strict cold-both protocol, the 70/10/20 ratios are applied independently to drug identifiers and exact target-sequence groups. Only pairs for which both entities belong to the same train, validation, or test block are retained; cross-block combinations are excluded to prevent either entity from crossing a partition. Consequently, the retained pair counts are 15,190/264/1,144, and the pair-level proportions are not 70/10/20. Of the 30,056 Davis pairs, 13,458 cross-block pairs are not used in this cold-both evaluation.

To quantify residual target relatedness after exact-sequence grouping, we used Biopython `Align.PairwiseAligner` in global mode (match $= 1$, mismatch $= 0$, gap opening $= -1$, gap extension $= -0.5$) to align each exact-sequence-grouped cold-target test sequence against every training sequence. Identity is the number of exact-residue matches divided by full alignment length; the maximum is retained for each test target. We also implemented leakage-aware diagnostic controls on this same exact-sequence-grouped cold-target partition. A drug-only control uses non-chiral radius-2, 1024-bit Morgan fingerprints in a logistic-regression classifier; a drug-promiscuity control uses each drug's positive prevalence estimated only from training targets; and a label-shuffled drug-only control permutes training labels while leaving test labels unchanged. Because every target identifier in this split is unseen, a target-specific degree computed from its full label column would leak test labels; the only target-only control that avoids that leakage assigns the training positive prevalence uniformly. These controls are diagnostic rather than capacity-matched replacements for BioInteract.

4.3 Evaluation Metrics

For binary classification under severe class imbalance, we report AUROC and AUPRC as primary metrics—AUPRC is particularly informative because it measures positive-class precision across all recall levels. For every reported protocol, F1, precision, and recall use a decision threshold selected once on the corresponding validation precision–recall curve; AUROC and AUPRC remain threshold-free.

4.4 Implementation

BioInteract is implemented in Python 3.10.14 using PyTorch 2.4.0+cu121 and PyTorch Geometric 2.6.1.[34] Protein embeddings are extracted using ESM-2[11] (`esm2_t30_150M_UR50D`, 640 dimensions; `fair-esm` 2.0.0). Molecular featurisation is performed with RDKit 2025.03.6.[35] Revision diagnostics additionally use Biopython 1.87 and RapidFuzz 3.14.5. Training employs automatic mixed precision (AMP) for memory efficiency on a single NVIDIA GPU. Final checkpoint evaluation is run under `torch.inference_mode()` in full precision, using each checkpoint's embedded architecture configuration.

5 Results

5.1 Comparison with Baseline Methods

We compare BioInteract against six baselines spanning sequence, graph, attention, transformer, and bilinear interaction models: DeepDTA,[5] GraphDTA,[6] AttentionDTA,[22] MolTrans,[7] TransformerCPI,[21] and DrugBAN.[8] Table 1 summarises representational contrasts with BioInteract without assigning causal credit to any single component. The six baseline values are reproduced from the comparison table in the originally submitted manuscript to document that submitted benchmark. This comparison is not a controlled state-of-the-art comparison: the baselines do not use a matched frozen ESM-2 representation and the methods differ in pretraining and original implementation choices. We therefore restrict comparative claims to the listed methods and report diagnostic controls separately.

Table 1. Representational scope of the selected historical baselines. These contrasts describe model inputs and interaction modules; they are not controlled component tests.

Baseline	Representational contrast with BioInteract
DeepDTA	Sequence-CNN drug and target encoders without residue-level ESM-2 input
GraphDTA	Molecular-graph drug encoder with a CNN target encoder
AttentionDTA	Self-attention-based DTI representation
MolTrans	Transformer substructure interaction representation
TransformerCPI	Transformer compound–protein representation
DrugBAN	Bilinear-attention interaction representation

Table 2 and Fig. 1 provide contextual comparison with the previously compiled benchmark table. The current-release BioInteract reconstruction yields random-split AUROC/AUPRC of 0.904/0.560. These values do not separate the effect of ESM-2 pretraining from the remaining architecture and are not presented as a fair comparison or a rank claim against contemporary ESM-based models.

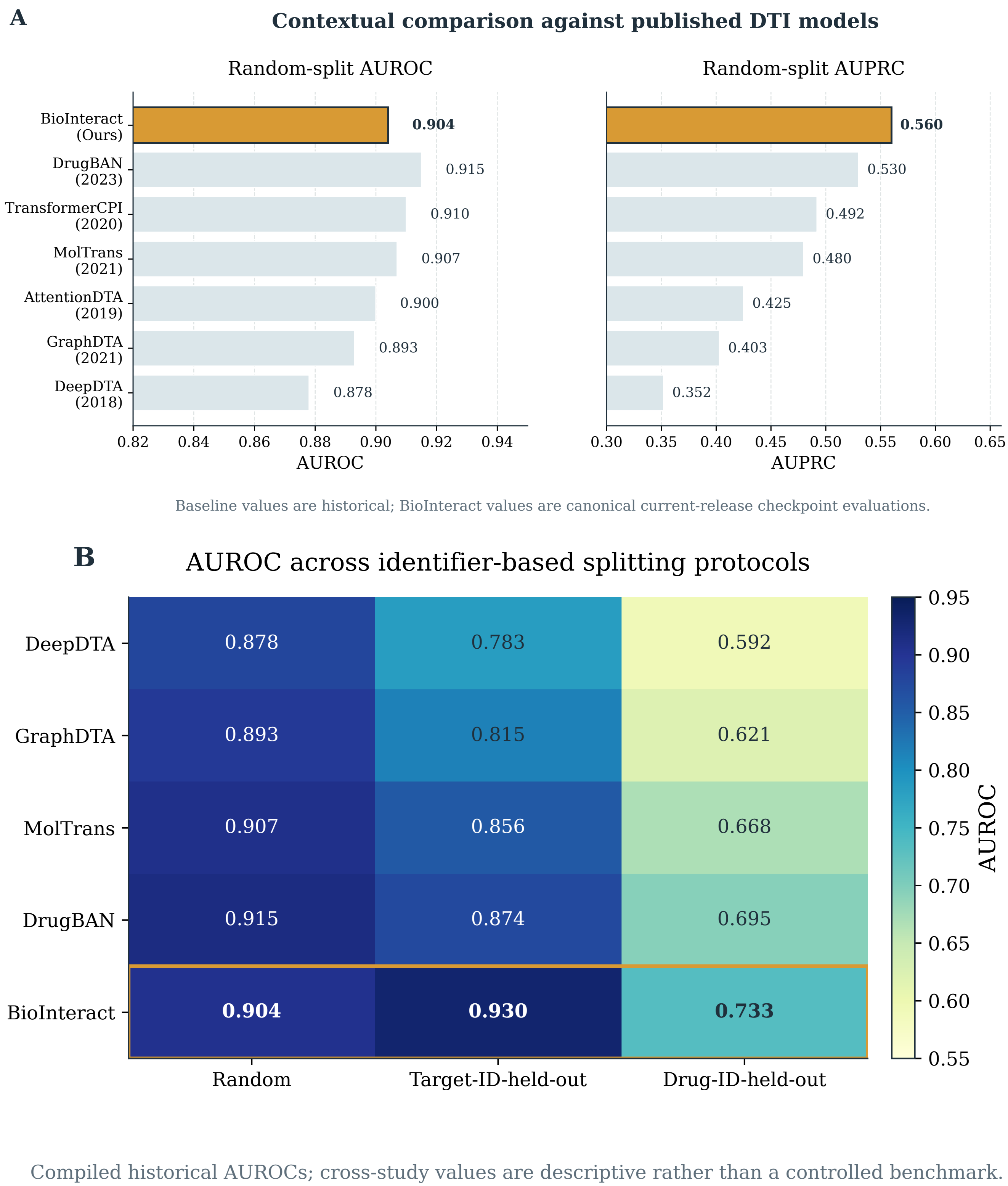

Figure 1. Baseline comparison. (A) Random-split AUROC and AUPRC for all methods. Baseline values are historical and unmatched; BioInteract values are from the canonical current-release checkpoint reconstruction. (B) AUROC across the three original entity-based splitting protocols. The Target-ID-held-out column is shown for transparency, but is not interpreted as sequence-novel generalisation because exact duplicate sequences crossed the original training and test partitions.

Table 2. Baseline values reproduced from the original submitted Davis comparison under three identifier-based splitting protocols. The entries are descriptive rather than a matched-pretraining or common-code benchmark.†

		Random		Target-ID-held-out		Drug-ID-held-out	
Method	**Paradigm**	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
DeepDTA[5]	Seq-CNN	0.878	0.352	0.783	0.247	0.592	0.089
GraphDTA[6]	GNN+CNN	0.893	0.403	0.815	0.312	0.621	0.102
AttentionDTA[22]	Self-Attn	0.900	0.425	0.838	0.348	0.643	0.108
MolTrans[7]	Transformer	0.907	0.480	0.856	0.389	0.668	0.121
TransformerCPI[21]	Transformer	0.910	0.492	0.862	0.401	0.672	0.125
DrugBAN[8]	Bilinear	0.915	0.530	0.874	0.428	0.695	0.138
BioInteract	Cross-Attn	0.904	0.560	0.930	0.524	0.733	0.166

† Baseline values are reproduced from the comparison table in the originally submitted manuscript. Supporting Information Table S8 summarises their source and protocol context. BioInteract values come from the canonical full-precision checkpoint reconstruction recorded in the DOI archive.

The current-release Target-ID-held-out reconstruction yielded AUROC = 0.930. Our audit shows, however, that six exact-sequence groups crossed its training and test partitions. This score is consequently reported as an entity-held-out result, not as evidence of sequence-novel generalisation. Its difference from the random-split value may reflect relatedness within the kinase-focused Davis collection, including duplicated sequences, and is not used to infer broad protein-family transfer.

Under the Drug-ID-held-out protocol, BioInteract obtained AUROC = 0.733 in the current-release reconstruction. The maximum test-to-training Morgan Tanimoto similarity ranged from 0.196 to 0.538 (median 0.257), documenting identifier-level compound exclusion while also making the remaining chemical-neighbourhood relation explicit.

5.2 Architecture scope and withheld historical ablations

The final audit could not reproduce the historical full-model prediction files from the released checkpoints and inputs, and no checkpoint/prediction artefacts are available for the ablated variants. We therefore withdrew the numerical ablation table, figure, and component-effect claims rather than compare non-reconstructible historical runs with the current-release full-model reconstruction. The specified architecture includes graph encoding, residue encoding, a configured domain-label channel, and cross-attention, but this study does not make a causal performance attribution to any component.

5.3 BioInteract Performance Across Splits

Table 3 summarises detailed BioInteract metrics, and Fig. 2 visualises the canonical current-release and strict-split point estimates. Full split sizes, positive counts, thresholds, and separately computed bootstrap intervals are provided in Tables S5–S7 of the Supporting Information.

Table 3. Threshold-free ranking metrics for BioInteract across the three original splits. "Target-ID-held-out" denotes an identifier-level partition that contained exact duplicate sequences across training and test.

Split	**AUROC**	**AUPRC**
Random	0.904	0.560
Target-ID-held-out	**0.930**	**0.524**
Drug-ID-held-out	0.733	0.166

Table 4. Strict within-Davis cold-start evaluation after grouping target identifiers with identical amino-acid sequences before partitioning. Both protocols use the original records, seed 42, and the same preprocessing and architecture. For cold-both, 70/10/20 is applied independently to drug IDs and target-sequence groups and only within-block pairs are retained. AUROC and AUPRC are threshold-free test metrics.

Strict protocol	Test pairs	Test positives	AUROC	AUPRC
Exact-sequence-grouped cold-target	5,984	208	0.873	0.383
Exact-sequence-grouped cold-both	1,144	38	0.552	0.038

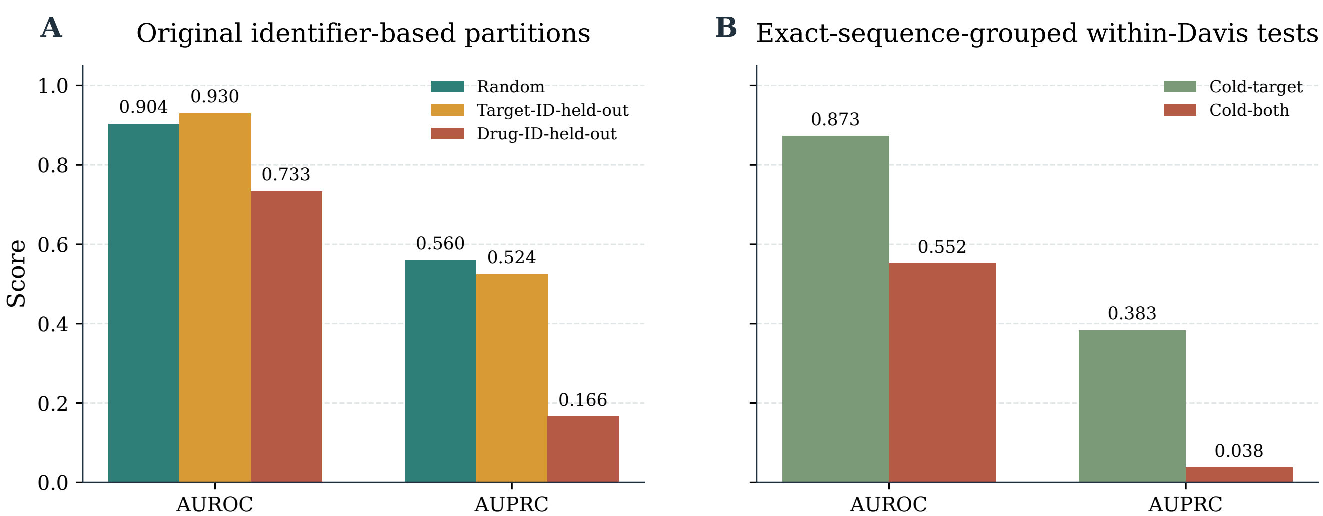

Figure 2. BioInteract performance and split audit. Panel A displays AUROC and AUPRC under the original random, Target-ID-held-out, and Drug-ID-held-out partitions from the canonical current-release checkpoint reconstruction. Panel B summarises the reviewer-requested exact-sequence-grouped cold-target and exact-sequence-grouped cold-both evaluations. The original Target-ID-held-out result is not interpreted as sequence-novel transfer.

The current-release Target-ID-held-out AUROC of 0.930 exceeds the random-split value, but this pattern is not interpreted as a biological transfer result. The maximum normalised Indel similarity between each test sequence and the original training targets ranges from 0.382 to 1.000 (median 0.579). The audit also found 18 duplicate-sequence groups involving 81 target identifiers in the full dataset and six groups crossing the original train/test boundary. We therefore report the exact-sequence-grouped results separately and limit conclusions to the observed Davis benchmark performance.

After grouping identical target sequences, the exact-sequence-grouped cold-target test gives AUROC = 0.873 and AUPRC = 0.383 (Table 4). Requiring both unseen drug identifiers and exact-sequence-grouped targets is substantially harder (AUROC = 0.552, AUPRC = 0.038); this test contains 38 positive pairs. These values are reported as an informative within-Davis stress test, not as a claim of out-of-family or structure-level generalisation. Threshold-dependent metrics for these small positive sets are secondary and use a threshold selected only on the corresponding validation set.

The cold-both train, validation, and test blocks retain 15,190, 264, and 1,144 pairs, respectively; 13,458 cross-block pairs are excluded by the double-held-out construction. The validation block contains only 10 positives. At its validation-selected threshold (0.936), the cold-both test yields F1, precision, and recall of 0.000, underscoring that its near-random AUROC and low AUPRC do not support a positive generalisation claim. Full counts and threshold-dependent metrics are reported in Table S7.

Table 5. Leakage-aware diagnostic controls on the exact-sequence-grouped cold-target test. All controls use only training-partition labels. They quantify available drug-side signal; they are not capacity-matched substitutes for BioInteract.

Model or control	AUROC	AUPRC
BioInteract (exact-sequence-grouped cold-target)	0.873	0.383
Drug-only Morgan logistic	0.764	0.164
Drug-promiscuity prior	0.764	0.163
Target-only constant prior	0.500	0.035
Label-shuffled drug-only	0.407	0.029

The drug-only and drug-promiscuity controls reach AUROC values near 0.764, showing that ligand-side activity or promiscuity signal contributes materially to this within-kinome task (Table 5). BioInteract exceeds these diagnostic controls, but the comparison does not prove that its residue attributions recover physical interactions. The label-shuffled control is near or below the positive-prevalence baseline, confirming that the drug-only signal depends on the observed training labels rather than fingerprint ordering.

For the exact-sequence-grouped cold-target test, maximum global sequence identity to the training set ranges from 0.231 to 0.867 (median 0.494): 31 of 88 held-out targets have a maximum identity below 0.40, 28 fall between 0.40 and 0.60, 24 between 0.60 and 0.80, and 5 are at least 0.80. BioInteract AUROC/AUPRC by these bins are 0.806/0.305, 0.908/0.456, 0.908/0.466, and 0.862/0.121, respectively. The final bin contains only three positive pairs, so its AUPRC is especially unstable. This non-monotonic, small-sample diagnostic does not establish robust transfer to homologously distant targets, but makes the relatedness distribution visible.

Under the Drug-ID-held-out split, BioInteract achieves AUROC = 0.733 in the canonical current-release reconstruction. The fingerprint audit is included to clarify that Drug-ID-held-out means unseen compound identifiers, not an externally validated scaffold-hopping or out-of-distribution guarantee.

5.4 Training Dynamics

Fig. 4 reports the single-seed validation-AUROC and training-loss trajectories for the archived protocols, with early stopping between epochs 32 and 47. These plots describe the observed optimisation paths only; they do not establish an absence of overfitting, quantify validation-loss behaviour, or estimate repeated-seed variability. They also do not isolate the contribution of any one regularisation choice, and the final reported runs use fixed molecular graphs.

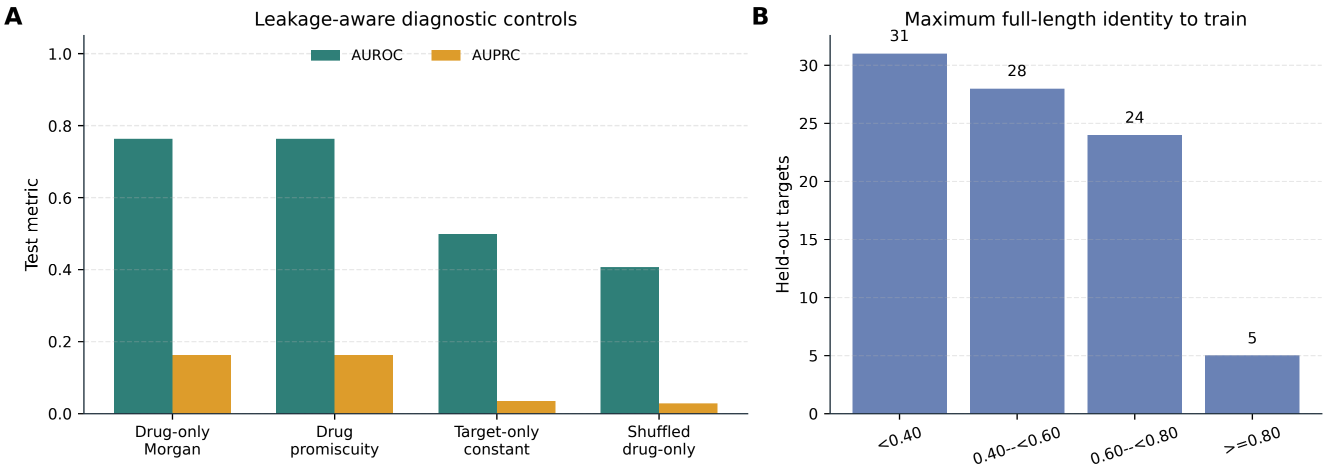

Figure 3. **Diagnostic controls and target-relatedness audit.** Panel A: leakage-aware controls on the exact-sequence-grouped cold-target test. Drug-only and drug-promiscuity scores use no target representation; the target-only constant is the only target-only control that does not use held-out target labels. Panel B: number of held-out targets by maximum full-length global sequence identity to any training target.

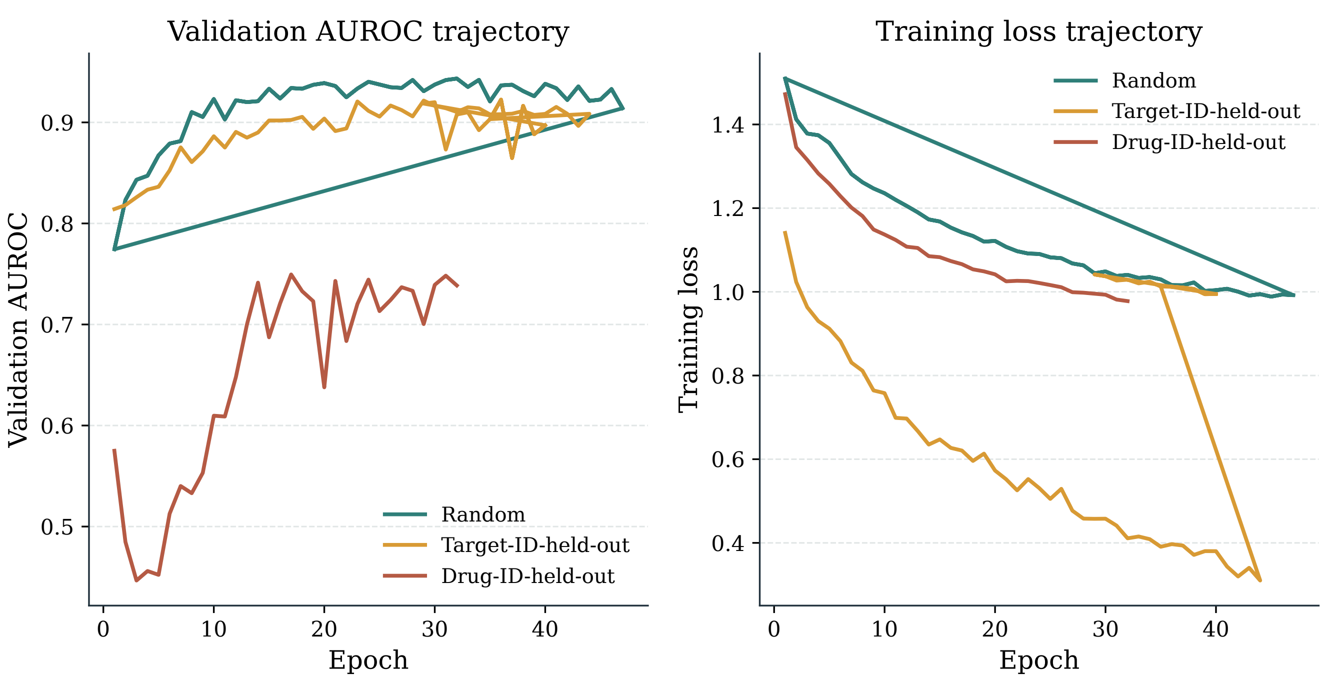

Figure 4. **Training dynamics across splitting protocols.** Panel A: validation AUROC over training epochs. Panel B: training loss curves. The trajectories describe the observed optimisation under the reported settings and do not establish an absence of overfitting, validation-loss behaviour, or repeated-seed variability. They also do not isolate the contribution of individual regularisation choices. Dashed vertical lines indicate early-stopping points.

5.5 Interpretability: Cross-Attention Analysis

All interpretability figures and tables are derived directly from the saved case-level outputs in the project interpretability-results directory, including the report JSON and the corresponding heatmap, profile, and Grad-CAM renderings. Accordingly, we report the model's actual residue-level attention and atom-level Grad-CAM outputs rather than schematic placeholders. These are attribution outputs of the trained model, not independently validated residue contacts or functional-group assignments.

5.5.1 Global Attention Sparsity

Analysis of cross-attention patterns across all 1,506 positive Davis drug–target pairs reveals a global attention sparsity of 99.2 % (Table 6, Fig. 5) under the stated normalisation threshold: fewer than 1 % of residues receive high relative attention. This describes concentration of the model attribution; it does not demonstrate the size or location of a physical binding interface.

Table 6. Global cross-attention statistics computed across all 1,506 positive Davis pairs using `checkpoints/best.pt`. The values quantify the concentration of the model's pair-normalised residue-attribution scores; they do not provide structural validation of a binding pocket.

Metric	Value
Mean residue attention	0.0044
Median residue attention	0.0003
Top-1% threshold	0.0572
Top-5% threshold	0.0036
Fraction of residues with normalised attribution < 0.1 of the pair-specific maximum	99.2%
Samples analysed	1,506

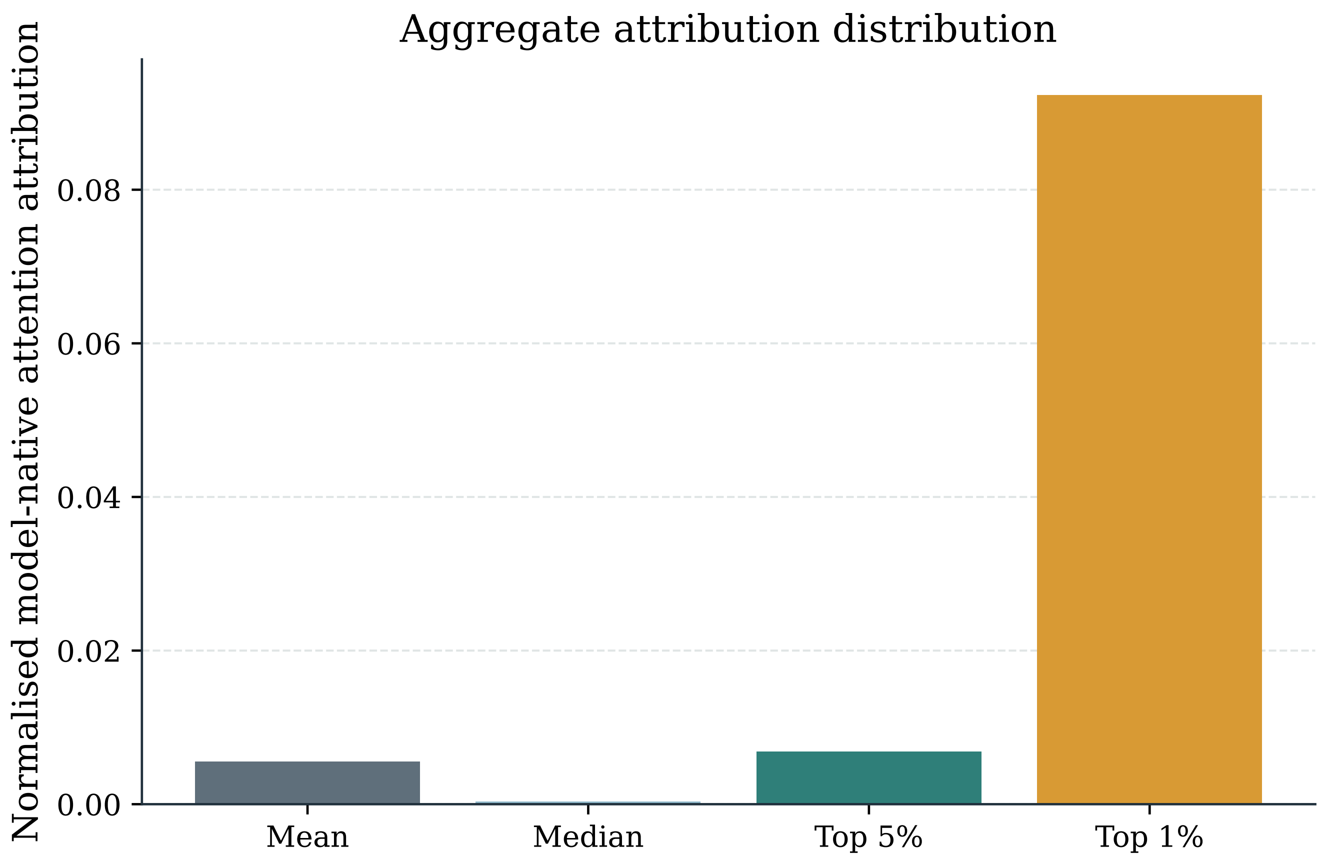

Figure 5. Attention sparsity analysis. (A) Distribution of pair-normalised per-residue attention scores across all 1,506 positive Davis pairs, displaying a heavy-tailed attribution distribution. (B) Cumulative attention as a function of residue percentile, showing that the top 1 % of residues capture a disproportionate fraction of total model attention. The 99.2 % value is an attribution concentration statistic, not a claim about a contiguous protein-surface region.

Although softmax attention is dense before normalisation, the observed concentration arises after training on the limited and imbalanced Davis data. It may reflect learned regularities, data composition, or both; it is therefore not presented as independent evidence that the model has recovered molecular recognition principles.

5.5.2 Audit of Nominal ABL1 Variant Inputs

A post-hoc audit of the exact Davis model inputs identified 15 target records whose nominal names begin with ABL1, including the wild-type and the named F317I, F317L, M351T, and E255K records. All 15 records have the same 1,167-residue amino-acid sequence and SHA-256 digest in `target_sequences.csv` (Supporting Information, Table S10). Several nominal ABL1 variant identifiers in the Davis records used here are associated with identical amino-acid sequences. Consequently, the present model inputs do not encode the corresponding mutation-specific sequence changes, and BioInteract cannot be evaluated for mutation-specific or resistance-specific behaviour using these records. We therefore withdraw the previous mutant-comparison analysis and draw no resistance-related conclusion.

5.6 Case Study: Functional-Group Attribution

The saved Grad-CAM output for the non-mutant EGFR–D0014 Davis pair provides a compact example of atom-level scores aggregated to functional groups (Table 7). The entries are model-native attribution scores rather than measured contacts or interaction types.

Table 7. Model-native functional-group attribution for the archived EGFR–D0014 Davis case. Scores are normalised atom-level Grad-CAM values aggregated by functional group; they do not identify physical contacts.

Functional Group	Importance
Carbonyl	0.431
Amide	0.389
Ether	0.136
Halogen	0.134
Amino	0.090
Aromatic ring	0.000
Heterocyclic N	0.000

5.7 Cross-Case Functional-Group Attribution Variation

The two non-mutant saved cases differ markedly in their classifier scores and attribution profiles (Table 8). EGFR–D0014 has a high-affinity-class score of 0.995 and its largest aggregated groups are carbonyl (0.431) and amide (0.389). BRAF–D0023 has a score of 0.099 and only small saved values for aromatic ring (0.059) and hydroxyl (0.000), despite its archived positive affinity of 0.19 nM. We therefore treat BRAF–D0023 as a low-scoring failure-mode example for an experimentally positive pair. This is a descriptive limitation of model behaviour, not an explanation of target binding or an assessment of functional-group chemistry.

Table 8. Cross-case model-native functional-group attribution. Classifier scores and group values are saved-output summaries for the non-mutant EGFR–D0014 and BRAF–D0023 Davis cases.

Case	Class score	Largest group	Score
EGFR–D0014	0.995	Carbonyl	0.431
BRAF–D0023	0.099	Aromatic ring	0.059

6 Discussion

6.1 From Prediction Accuracy to Inspectable Hypotheses

BioInteract is intended to make model behaviour more inspectable than a purely scalar DTI prediction. Its attention matrices and Grad-CAM scores can generate hypotheses about which representations contributed to a prediction. Such hypotheses can subsequently be assessed with crystal structures, mutagenesis, or structure–activity relationship (SAR) experiments; the present study does not replace these measurements.

The contextual random-split comparison places the canonical BioInteract result below the historical DrugBAN AUROC and above its historical AUPRC; because the protocols and pretraining are unmatched, this is not a comparative performance claim. These Davis results do not show that BioInteract maps a physical binding pocket, assigns an interaction type to an atom pair, ranks a large compound library, or assesses treatment resistance.

6.2 The 99.2% Attention Sparsity

The observed 99.2 % attention sparsity is a property of the pair-normalised, residue-pooled attribution scores under the chosen threshold. Although it is compatible with a model focusing on a small subset of residues, it can also be affected by the kinase-focused and imbalanced training data. It should not be read as evidence that the highlighted residues constitute a true binding site.

6.3 Scope of Cold-Start Evaluation

The audit of the original Target-ID-held-out split identified exact duplicate sequences across the training and test partitions. We therefore withdraw the previous interpretation of the 0.930 AUROC as evidence of sequence-novel or protein-family-level transfer. The newly added exact-sequence-grouped cold-target and exact-sequence-grouped cold-both evaluations provide a stricter within-Davis stress test, but they still involve kinase proteins and do not demonstrate performance on unrelated protein families or pocket classes.

The Drug-ID-held-out result is likewise restricted to the chemical diversity of 68 kinase inhibitors. The observed Morgan-similarity distribution documents the nearest-neighbour relationship between held-out and training molecules, but it does not establish scaffold hopping or out-of-distribution robustness.

6.4 Diagnostic Controls and Remaining Dataset Bias

The exact-sequence-grouped cold-target diagnostic controls demonstrate that a target-independent drug representation retains substantial ranking signal in Davis. The drug-only Morgan and drug-promiscuity controls each achieve AUROC near 0.764, whereas BioInteract reaches 0.873 under the same exact-sequence-grouped partition. This improvement is compatible with the use of target information, but it neither excludes ligand-side activity bias nor establishes that the model has learned a molecular-recognition mechanism. We therefore interpret the control comparison as a boundary on the Davis result, not as causal evidence for any particular encoder or attribution map. The label-shuffled control confirms that the observed drug-only signal depends on the training labels; it is not a test of structural interaction recovery.

6.5 Functional-Group Attribution Without Structural Supervision

The Grad-CAM functional-group ranking is qualitatively compared with kinase SAR literature, but it remains a gradient-based attribution. In particular, the analysis does not determine whether a group forms a hydrogen bond, a pi-stacking interaction, a cation–pi interaction, or a van der Waals contact, nor whether an implicated protein residue contributes through its backbone or side chain. These questions require a structural complex or targeted experiments.[36]

6.6 Limitation of Nominal ABL1 Variant Records

The exact-input audit shows that the nominal ABL1 variant labels do not supply variant-specific sequence information to the present model. We therefore remove the former mutant-comparison figures, tables, and conclusions. The Davis records remain useful as labelled kinase interactions, but they cannot support an analysis of mutation effects or drug resistance in this study.

6.7 Future Directions

The Davis results motivate, but do not establish, several future directions. Future evaluation on chemically broader data and non-kinase targets is required before making a protein-family generalisation claim. Direct comparison with independent experimental evidence is also needed before attributions can be assigned a structural meaning. The present kinase-only data do not support a robustness claim beyond the analysed setting.

No overlap analysis between the Davis cases and PDBBind or LP-PDBBind, and no PLIP contact validation, was performed in this study. A future structural audit should use a non-overlapping benchmark of experimentally resolved complexes, predefine the atom–residue contact criterion, and compare model attributions with the resulting contact labels (for example, from PLIP) before structural meaning is assigned.[37]

The present implementation was audited on one NVIDIA GPU. Although the 2,442,083 trainable parameters do not increase with the number of training pairs, data-processing time, graph batching, ESM-2 embedding extraction and storage, and accelerator memory must be profiled at larger scale. Distributed training was not evaluated, and no claim is made that the current implementation scales linearly to BindingDB-sized data.

Evaluation on curated allosteric and cryptic-site benchmarks, with appropriate structural or dynamic context, is likewise required before BioInteract can be assessed for non-canonical drug–target interactions.

Extending BioInteract from binary interaction classification to continuous binding-affinity regression (pK_d) would provide the finer-grained potency estimates needed for lead optimisation, where accurate ranking of closely related analogues is as important as binary activity prediction.

7 Conclusion

We have presented BioInteract, a DTI-prediction framework that combines chemistry-aware molecular graph encoding, ESM-2/physicochemical residue representations, and bidirectional cross-attention. The model provides prediction scores together with inspectable model-native attention and Grad-CAM attributions.

The revision clarifies the evidence boundary of these outputs. The observed attention concentration and functional-group ranking describe model behaviour; neither is a validated three-dimensional mechanism or binding-contact prediction. We also distinguish Target-ID-held-out performance from exact-sequence-grouped cold-start evaluation after detecting duplicate sequences in the original target partition. Broader datasets and direct structural validation remain necessary for generalisation and mechanistic claims.

BioInteract is available as a publicly accessible interactive web interface at `https://huggingface.co/spaces/AI4deeperScience/BioInteract` for the custom-prediction workflow shown in Fig. 6. The repository README documents the command-line workflow for the available attribution outputs.

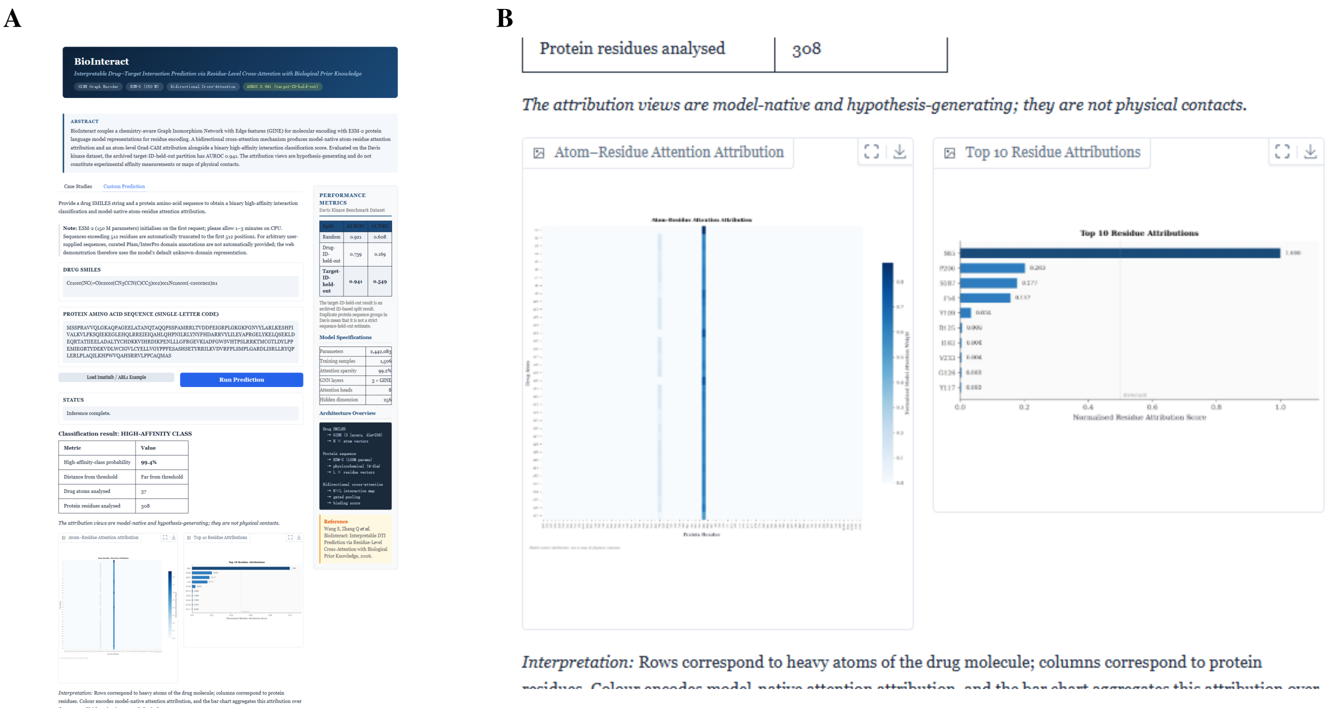

Figure 6. BioInteract interactive web server. Panel A shows the full screenshot of the publicly accessible interface at `https://huggingface.co/spaces/AI4deeperScience/BioInteract`. Panel B enlarges the attribution region from the same screenshot. The interface shows the custom-prediction input and prediction-result state for human Aurora kinase C (UniProt Q9UQB9). It displays the entered SMILES, protein sequence, high-affinity-class probability, and threshold-distance category. The enlarged lower portion displays the generated atom–residue attention map and the "Top 10 Residue Attributions" chart. These are model-native attribution views, not physical-contact or hotspot labels. Command-line instructions for generating the available attribution outputs are provided in the public repository.

BioInteract represents a step toward AI-assisted DTI prediction in which model scores are accompanied by transparent attribution outputs and explicit limits on their interpretation. Such outputs are most useful when combined with rigorous split auditing, external validation, and biochemical experiments.

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Supporting Information

The Supporting Information contains: model hyperparameters and feature details (Tables S1–S3); computational environment (Table S4); dataset and split statistics (Tables S5–S7); baseline scope (Table S8); atom features (Table S9); the Davis ABL1 nominal-variant input audit (Table S10); non-mutant EGFR and BRAF case summaries (Tables S11–S12); and complete model parameter counts (Table S13). This material is available free of charge.

Data Availability

All data used in this study are publicly available. The Davis kinase inhibitor dataset used for model training and evaluation is available from the original publication.[2] The immutable release containing the released checkpoints, canonical prediction CSVs, provenance hashes, strict-split artifacts, and reproduction scripts is archived at Zenodo, `https://doi.org/10.5281/zenodo.21429673`. The BioInteract source repository is `https://github.com/Liubo-520/bio-intelligence`. An interactive web server for the custom-prediction workflow is available at `https://huggingface.co/spaces/AI4deeperScience/BioInteract`; the README documents the command-line attribution workflow.

Code Availability

The BioInteract source code, pretrained model weights, configuration files, canonical metric-reconstruction scripts, and figure-generation scripts are publicly available at `https://github.com/Liubo-520/bio-intelligence` and as the immutable Zenodo release, `https://doi.org/10.5281/zenodo.21429673`. The interactive web application for the custom-prediction workflow is available at `https://huggingface.co/spaces/AI4deeperScience/BioInteract`.

Author Contributions

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Xucheng Liu: Investigation, Validation, Data curation.
Zixuan Wang: Software, Investigation, Data curation.
Jiangke Cheng: Resources, Validation, Visualisation.
Zhi Huang: Supervision, Writing – review & editing, Funding acquisition, Corresponding author.

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Declaration of Competing Interests

The authors declare no competing interests.

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